The Model That Doesn't Know

Out-of-Distribution Detection & Uncertainty Estimation
Comprehensive Learning Guide with Visualizations

CodeChef-VIT AI & Machine Learning Track
Promotional Recruitment 2026-27
Frozen ResNet-18 | Linear Probe | MSP | Energy | Mahalanobis

Part 1: The Problem

Standard deep learning classifiers are trained under a closed-world assumption: every test input will belong to one of the predefined training classes. When presented with an image from a category it has never seen, the model doesn't say 'I don't know.' Instead, softmax forces probabilities to sum to 1.0, so the model confidently assigns a label - often with >90% confidence - to something it was never trained on.

This is dangerous in safety-critical applications. A medical imaging system trained on lung diseases might confidently diagnose a broken wrist X-ray as 'healthy lung.' An autonomous vehicle might misidentify a new road sign. Confidently making an incorrect prediction is far more hazardous than admitting uncertainty.

Our project: Build a system that can say 'I don't know.'

We build a classifier that recognizes 5 known categories while deploying three independent statistical uncertainty detectors to flag unfamiliar inputs for rejection.

Not All 'Unknown' Is Equal

The key insight of this study is testing on TWO difficulty levels:

Near-OOD (Hard): Images that look similar to known classes. Dogs shown to a model trained on cats share fur, eyes, and body shapes. The model's features fire similarly.

Far-OOD (Easy): Images from a completely different domain. Fashion items (boots, shirts) shown to a natural-image classifier. Features won't overlap at all.

The gap between near-OOD and far-OOD performance is the central finding of the study.

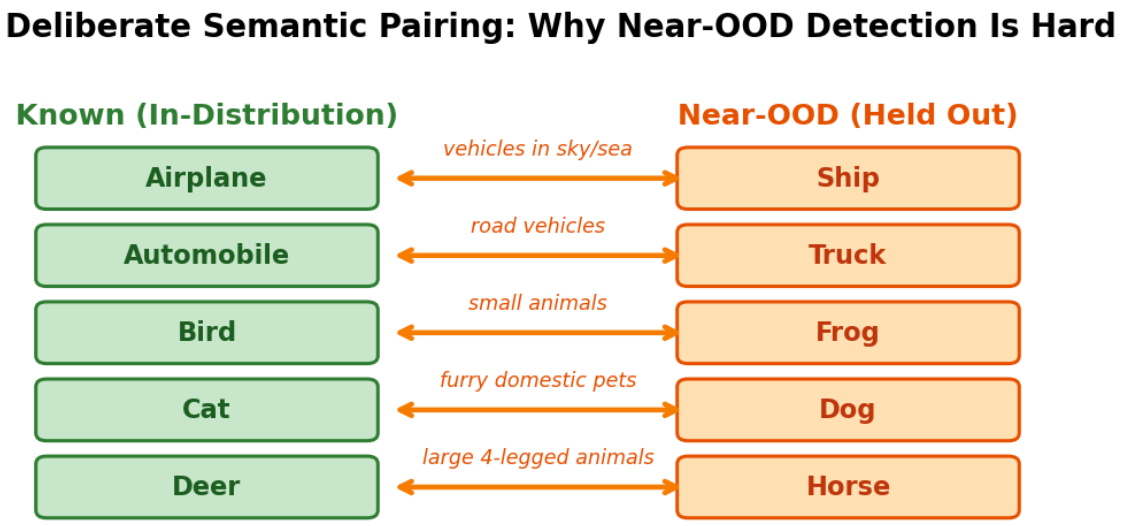


Fig 1: Deliberate semantic pairing - each unknown class has a known neighbor

Part 2: The Architecture & Pipeline

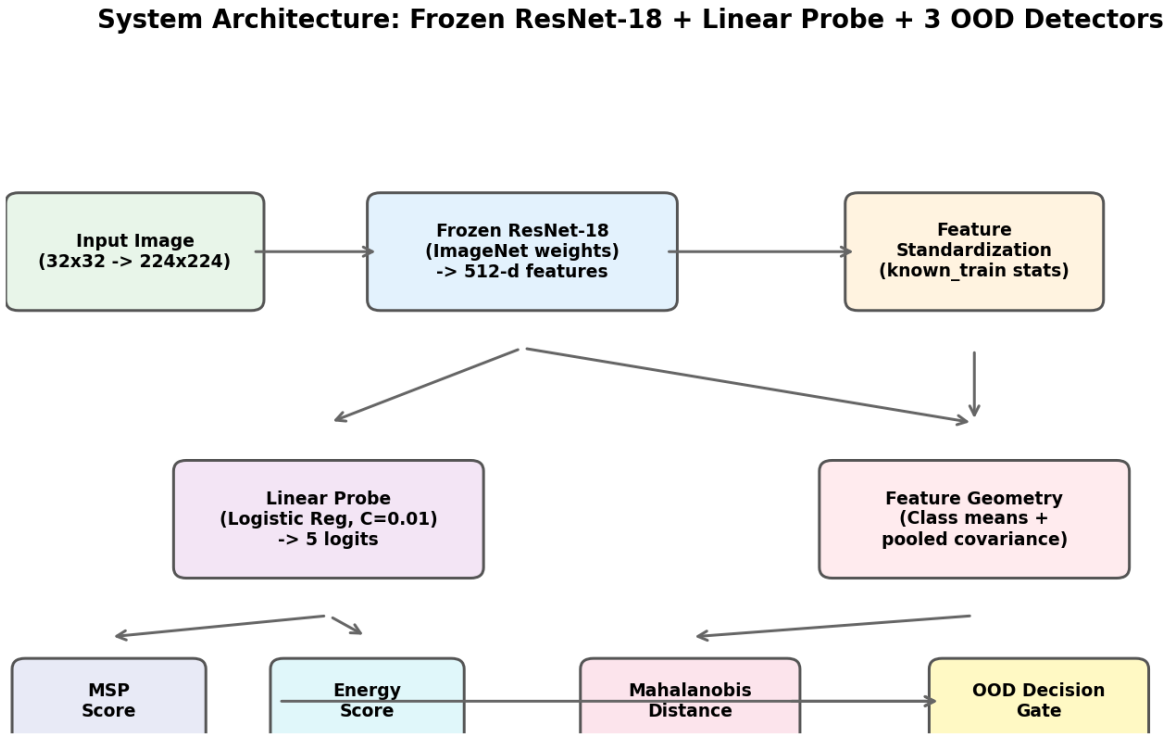


Fig 2: System architecture - Frozen ResNet-18 + Linear Probe + 3 OOD Detectors

The Data Split

We used CIFAR-10 (60,000 tiny 32x32 color images, 10 classes) split deliberately:

Known (In-Distribution): CIFAR-10 classes 0-4 (Airplane, Automobile, Bird, Cat, Deer)
Near-OOD: CIFAR-10 classes 5-9 (Dog, Frog, Horse, Ship, Truck)
Far-OOD: FashionMNIST test set (grayscale clothing items converted to 3-channel RGB)

Critical integrity rule: The held-out classes NEVER touch training, validation, temperature fitting, or scaler fitting. Any leakage would invalidate all results.

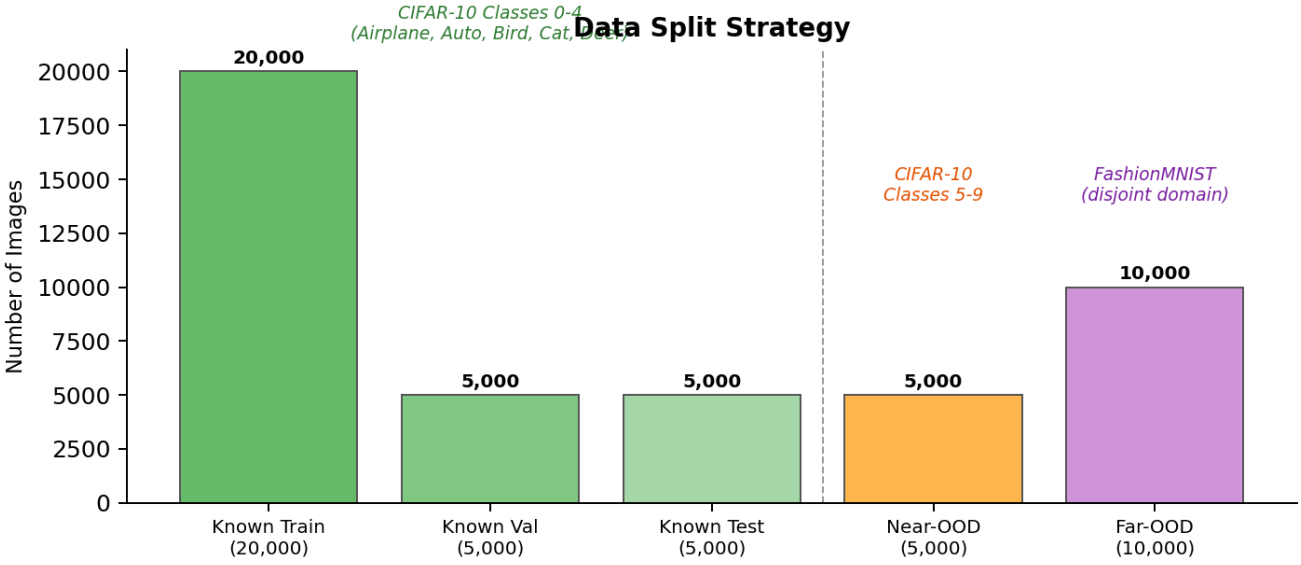


Fig 3: Data split sizes - balanced classes, strict separation

Frozen ResNet-18 + Linear Probe

Step 1 - Feature Extraction: We use ResNet-18 pre-trained on ImageNet with its final classification layer removed. All weights

are frozen (no gradients). Each image is resized to 224x224 and passed through, producing a 512-dimensional feature vector.

Step 2 - Feature Standardization: We compute the mean and standard deviation of these 512-d features on known_train ONLY, then apply z-score normalization to all splits.

Step 3 - Linear Classifier: A multinomial logistic regression (C=0.01, strongly regularized) is trained on the standardized known_train features. It outputs 5 raw logits.

Why freeze? Isolates the experiment to detection methods. Runs in minutes. Reproducible. Cost: ~92% accuracy instead of ~96%, but classification accuracy isn't our main metric.

Part 3: The Three OOD Scoring Methods

Convention: HIGHER score = MORE likely OOD (more unfamiliar)

Method 1: Maximum Softmax Probability (MSP)

Idea: If the model is uncertain, probabilities should be spread out, not concentrated.

Math: Softmax(z_i) = exp(z_i) / sum(exp(z_j))
MSP Score = -max_c Softmax(z)_c

Example: Confident output [0.02, 0.94, 0.01, 0.02, 0.01] => MSP = -0.94 (familiar)
Uncertain output [0.22, 0.18, 0.25, 0.15, 0.20] => MSP = -0.25 (unfamiliar)

Fatal flaw: Softmax is invariant to adding a constant to all logits. It only measures RELATIVE differences. An unfamiliar input with moderate logits can look just as confident as a familiar one with strong logits.

Method 2: Energy Score

Idea: Instead of just the winner, look at the TOTAL activation volume.

Math: E(x; T) = -T * log(sum(exp(z_j / T)))

Why it's better: Energy uses the logsumexp of ALL logits, capturing absolute magnitude. Familiar inputs trigger strong activations (large logits), producing very negative energy (e.g., -8.0). Unfamiliar inputs produce weaker activations, yielding less negative energy (e.g., -3.0).

Key insight: Softmax asks 'Which class wins?' (relative). Energy asks 'How strongly does the model react overall?' (absolute). That's why a 94%-confident cat can be rejected by Energy - the overall logit magnitudes are moderate.

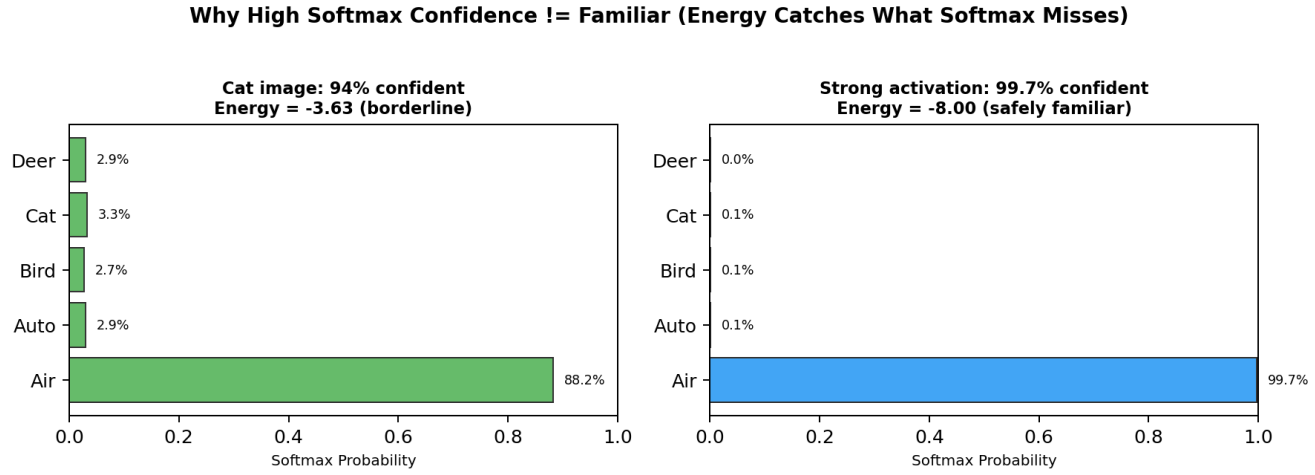


Fig 4: Why 94% softmax confidence != familiar - Energy catches what Softmax misses

Method 3: Mahalanobis Distance

Idea: Forget classifier outputs entirely. Look at WHERE the image sits in feature space.

Math: D_M(x) = min_c sqrt((f(x) - mu_c)^T * Sigma^-1 * (f(x) - mu_c))

During training, each class forms a 'cluster' in the 512-d feature space. Mahalanobis measures the distance from a new image to the nearest cluster center, corrected for how spread out each cluster is.

Why it's different: MSP and Energy use logits (classifier outputs). Mahalanobis uses raw feature geometry. It fails differently -

which makes the 3-method comparison informative.

Implementation: Cholesky decomposition (cho_factor/cho_solve) instead of direct matrix inversion. Ridge regularization (lambda=0.001) guarantees positive definiteness.

Part 4: Experimental Results

Classification Accuracy: 92.02%

On 5,000 test images across 5 known classes, the model achieves 92.02% accuracy. This is within the expected range (88-97%) for a frozen ResNet-18 linear probe on upscaled CIFAR-10. Not state-of-the-art, but classification accuracy isn't our goal - OOD detection is.

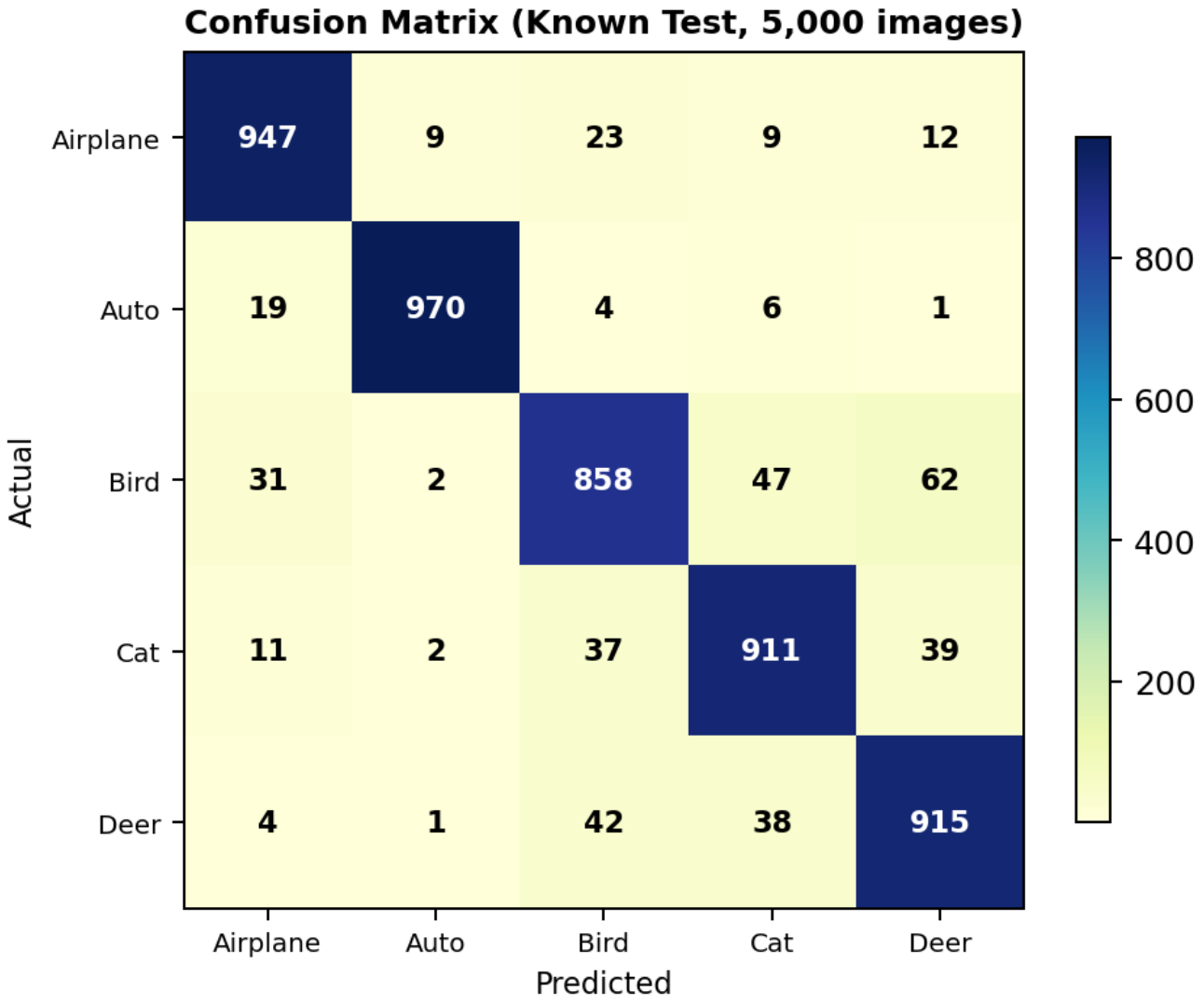


Fig 5: Confusion matrix - plausible errors (bird/deer, cat/deer)

OOD Detection Performance

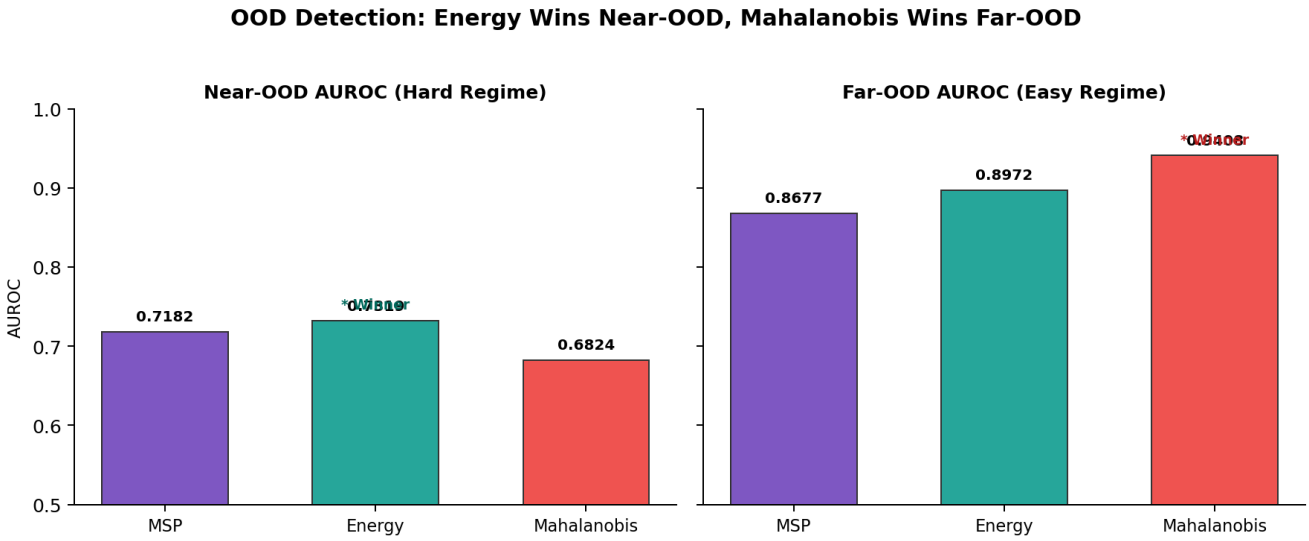


Fig 6: AUROC comparison - Energy wins Near-OOD, Mahalanobis wins Far-OOD

KEY FINDINGS:

1. Energy beats MSP on near-OOD (0.7319 vs 0.7182) - absolute > relative logit measurement
2. Mahalanobis wins far-OOD (0.9408) - geometric distance excels when domains are disjoint
3. Mahalanobis loses near-OOD (0.6824) - overlapping feature clusters cannot be separated
4. Near-OOD is genuinely hard (FPR@95%: 73-81%) - this is expected per the literature
5. Far-OOD is easy (FPR@95%: 32-40%) - different domain = clean separation

The Threshold Trade-Off

After computing scores, we need a decision threshold: above this energy, reject.

There is no free lunch:

Permissive (-2.50): Accepts most known images, but lets some impostors through

Balanced (-3.60): Derived so <=10% of known inputs are falsely rejected

Strict (-8.13): Catches 95% of near-OOD, but rejects ~73% of known inputs

This trade-off is fundamental and unavoidable - it is the ROC curve.

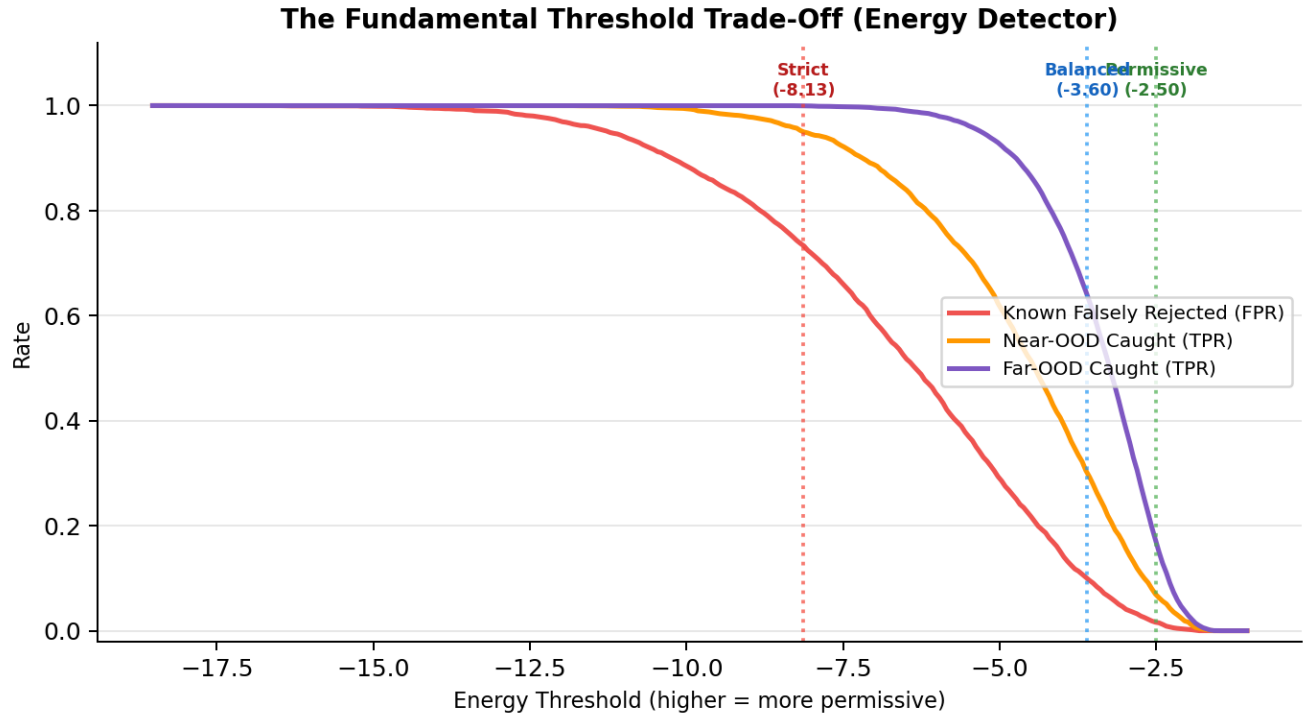


Fig 7: The threshold trade-off - no single threshold works for everything

Part 5: Published Diagnostic Figures

Score Distributions

Histograms showing how Known (green), Near-OOD (orange), and Far-OOD (purple) scores are distributed. Notice the heavy overlap between Known and Near-OOD - that is why near-OOD detection is fundamentally hard. Far-OOD separates cleanly.

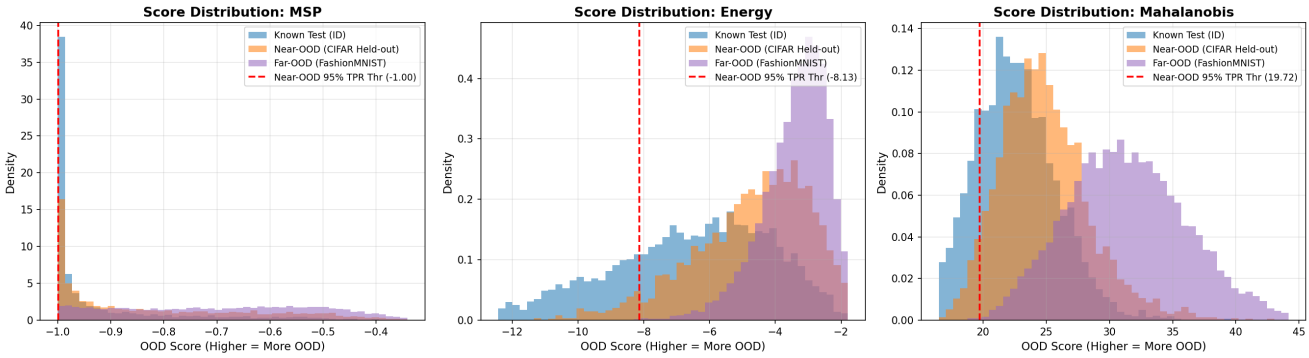


Fig 8: Score distributions - near-OOD overlaps known; far-OOD separates cleanly

Risk-Coverage Curve

This answers: 'If the model refuses its least confident predictions, how much does accuracy improve?' At full coverage (accept everything): 92% accuracy. At 50% coverage (only accept top-half most confident): accuracy rises above 99%. This proves the value proposition: a model that can say 'I don't know' is more reliable.

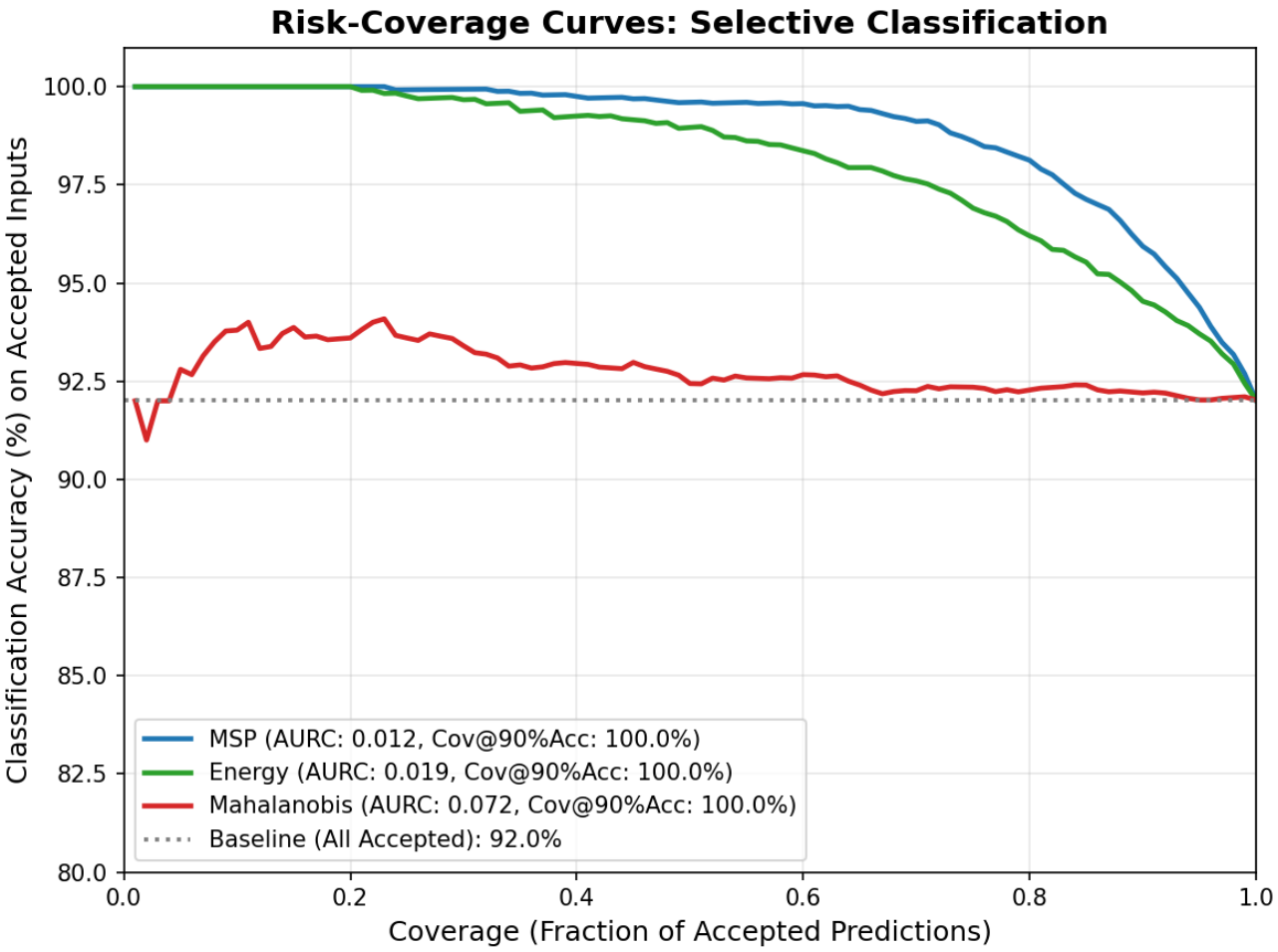


Fig 9: Selective classification - rejecting uncertain predictions improves accuracy 9x

Attractor Analysis

Some known classes act as magnets for unknown inputs. Cat (AUROC 0.6789) and Deer (AUROC 0.6825) are the weakest - they attract dogs and horses respectively. Automobile (0.8720) is surprisingly strong against trucks.

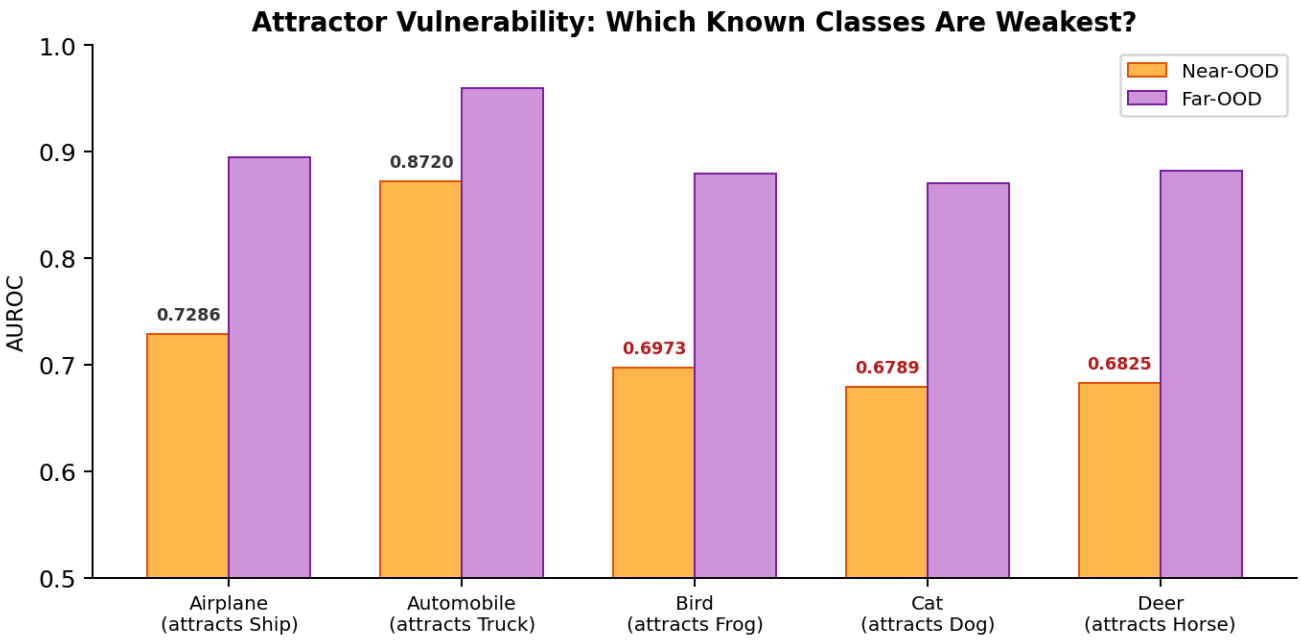


Fig 10: Per-class AUROC - Cat and Deer are the most vulnerable attractors

Failure Gallery

OOD images that FOOLED the detector at 95% TPR. Orange border = Near-OOD, Purple = Far-OOD. Each tile shows predicted class and confidence. Dogs predicted as cats, horses as deer - the semantic attractor effect visualized.



Fig 11: Failure gallery - falsely accepted OOD inputs at 95% TPR threshold

Part 6: Key Decisions & Known Weaknesses

Design Decisions

D1: Frozen ResNet-18 instead of fine-tuning

Why: Isolates the experiment to detection methods. Cost: ~92% vs ~96% accuracy. Acceptable.

D2: CIFAR-10 split instead of custom dataset

Why: Same image statistics, only class identity differs. No confounding domain shift.

D3: C = 0.01 (strong regularization)

Why: Prevents large logit magnitudes. Natural calibration (ECE 0.77%). Cost: slightly lower accuracy.

D4: Three methods instead of one

Why: They fail differently. MSP=baseline, Energy=best logit method, Mahalanobis=feature geometry.

Known Weaknesses (Be Honest)

W1: Near-OOD detection is weak - AUROC 0.73 is only marginally better than random. FPR@95% of 73% means rejecting 73% of known inputs to catch 95% of impostors. This is expected and reported honestly.

W2: Custom image upload is unreliable - CIFAR-10 features (32x32) don't match real-world high-resolution photos. This is a domain gap, not a bug.

W3: Single backbone, single seed - no ensemble uncertainty or variance estimation.

W4: Temperature scaling changes little - ECE goes from 0.77% to 0.79%. The honest finding: strongly regularized linear probes are naturally well-calibrated.

Part 7: Vocabulary Quick Reference

OOD	An input the model has never seen during training
Near-OOD	Unknown input that looks similar to known classes (dog vs cat)
Far-OOD	Unknown input from a completely different domain (boot vs cat)
AUROC	Probability that a random OOD scores higher than a random known. 0.5=random, 1.0=perfect
FPR@95% TPR	To catch 95% of unknowns, what fraction of known images are accidentally rejected
Softmax	Converts logits to probabilities summing to 1.0 - measures relative class preference
Energy Score	Measures absolute activation strength - how strongly the model reacts overall
Mahalanobis	Distance from an image's features to the nearest known class cluster
ECE	Average gap between predicted confidence and actual accuracy
Temperature T	Divide logits by T to fix overconfidence. Our T=0.946 (already well-calibrated)
Logit	Raw pre-softmax number for each class. Higher = model thinks more likely
ROC Curve	Plot of TPR vs FPR as threshold sweeps. Hugs top-left = good
Linear Probe	Simple logistic regression on frozen features - no neural network training
Semantic Attractor	A known class that 'pulls in' unknown inputs (Cat attracts Dogs)

Links & Reference

GitHub Repository:
https://github.com/Satyakarma1/the-model-that-doesnt-know

Live Streamlit Dashboard:
https://satyakarma1-the-model-that-doesnt-know-app-zd4ljo.streamlit.app/